



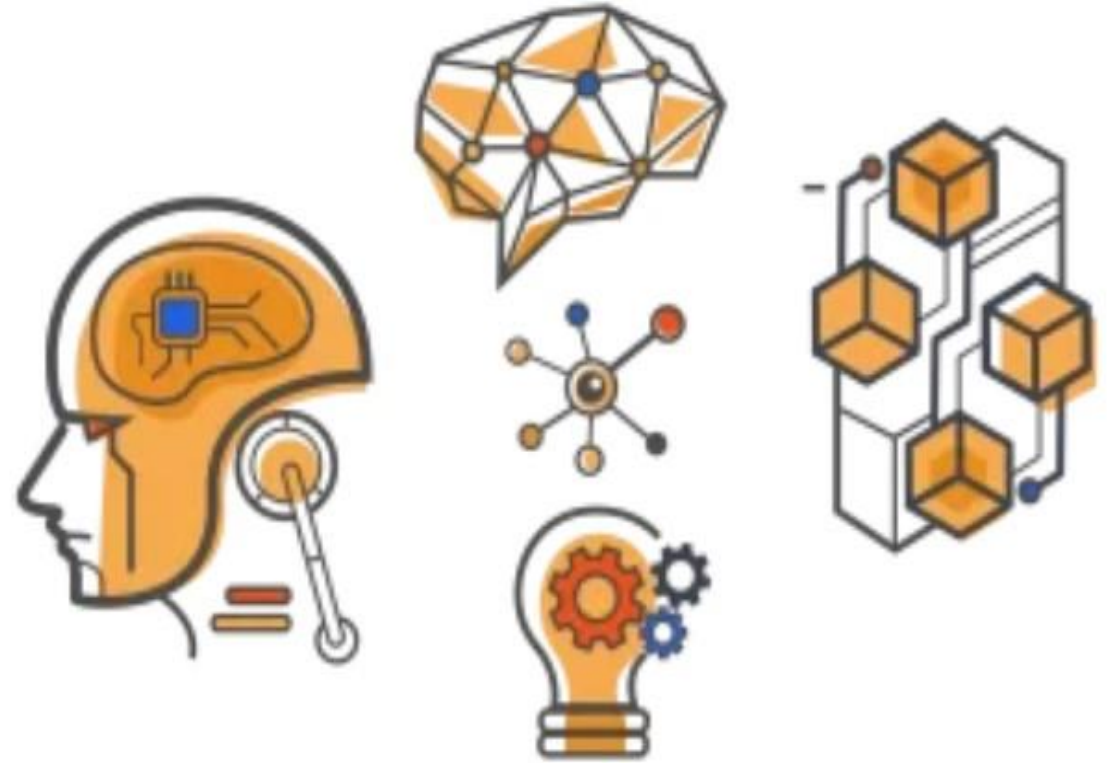
COSC-E4 CS ELECTIVE 4 (MACHINE LEARNING)

by:

Montaigne G. Molejon, MSIT

LESSON 1:

Introduction to Machine Learning





Learning Objectives

Upon completion of this lesson, you should be able to:

- Understand the basic concepts of Machine Learning (ML).
- Define the importance of Machine Learning (ML).
- Apply the process of Machine Learning (ML).



Introduction

What is **Learning**?

- *"A change in human disposition or capability that persists over a period of time and is not simply ascribable to **processes of growth**."*

*- From **The Conditions of Learning** by Robert Gagne*

- *"Learning is the relatively **permanent change in a person's knowledge or behavior due to experience**. This definition has three components:*

- 1) *The duration of the change is **long-term** rather than short-term;*
- 2) *The **locus** of the change is the content and structure of knowledge in memory or the behavior of the learner;*
- 3) *And the **cause of the change is the learner's experience** in the environment rather than fatigue, motivation, drugs, physical condition or physiologic intervention".*

*- From **Learning in Encyclopedia of Educational Research**, Richard E. Mayer*



Introduction (Cont'd)

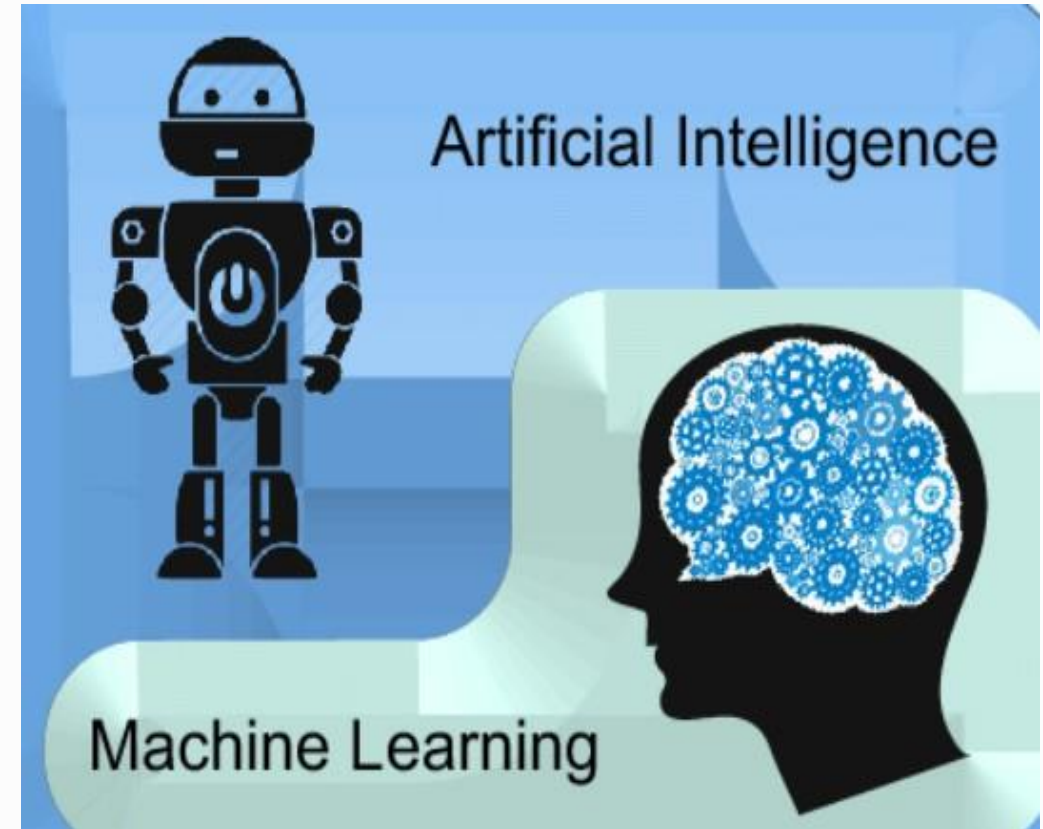
What is **Learning**?

- *"We define learning as the **transformative process** of taking in information that, when internalized and mixed with what we have experienced—changes what we know and builds on what we do. It's based on **input, process, and reflection**. It is what changes us."*
 - *From **The New Social Learning** by Tony Bingham and Marcia Conner*
- *"A process that leads to change, which occurs as a result of experience and increases the potential of improved performance and future learning."*
 - *From **How Learning Works: Seven Research-Based Principles for Smart Teaching** by Susan Ambrose, et al.*

Introduction (Cont'd)

What is Machine Learning?

- A branch of **Artificial Intelligence (AI)** that allows software applications to become more **accurate** at **predicting outcomes** without being explicitly programmed to do so.
- Machine learning algorithms use **historical data** as input to predict new output values.



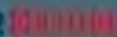
Brief History of Machine Learning

1950

- **Alan Turing** published a paper answering the question "*Can Machine Think?*".

I propose to consider
the question, 'Can
machines think?'

- Alan Turing

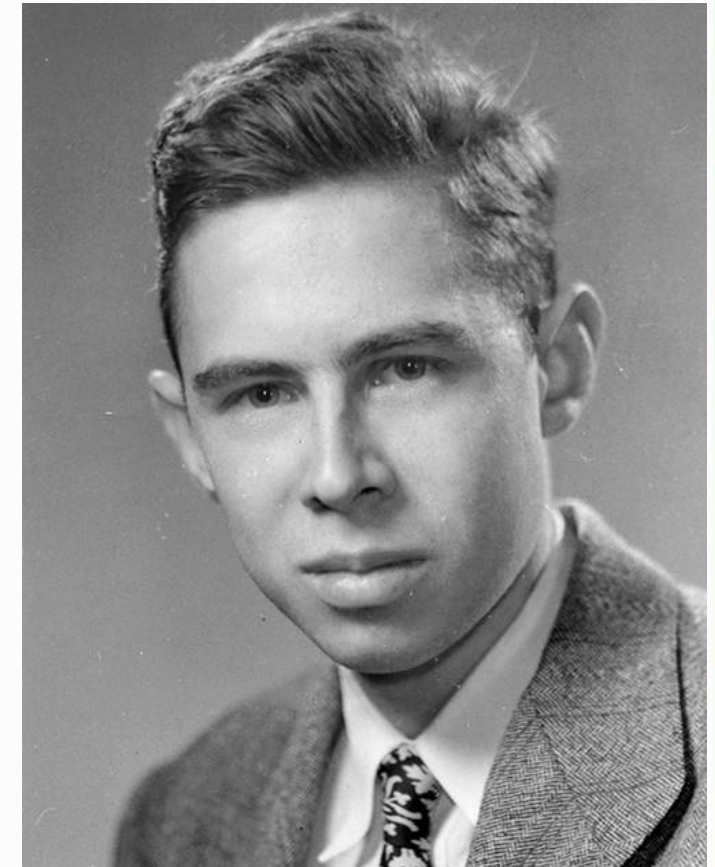
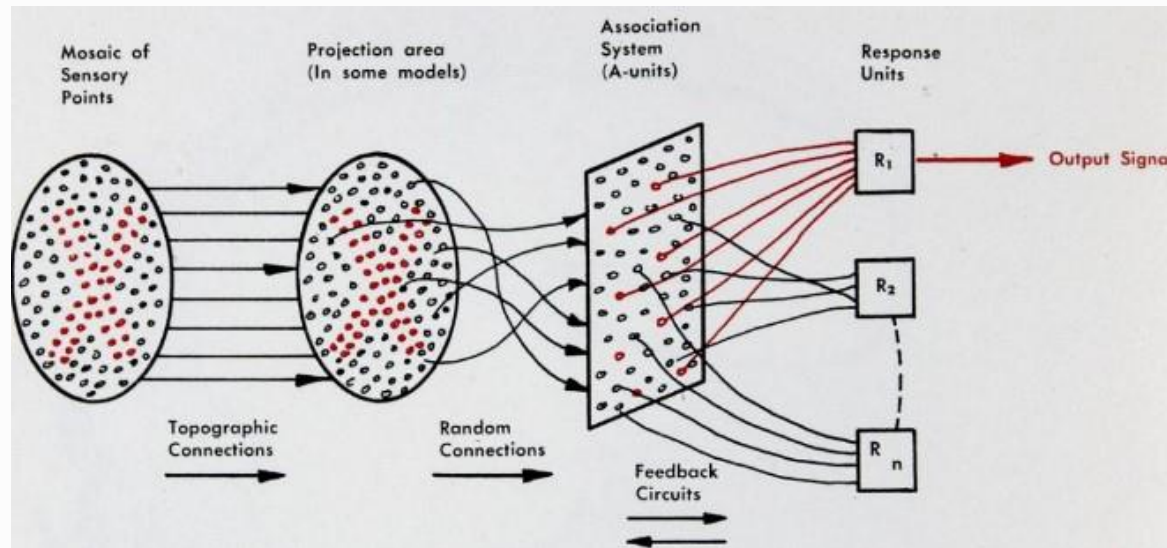
SPRINGER 
On This Day



Brief History of Machine Learning (Cont'd)

1957

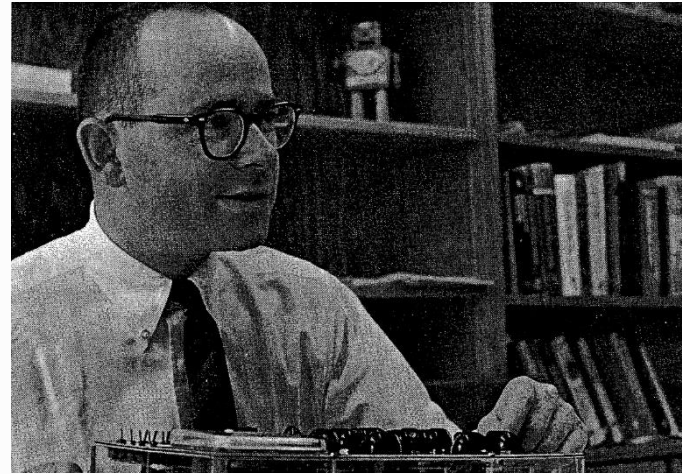
- First neural network that was designed by **Mr. Frank Rosenblatt**. Also called as **Perceptron Model**.



Brief History of Machine Learning (Cont'd)

1959

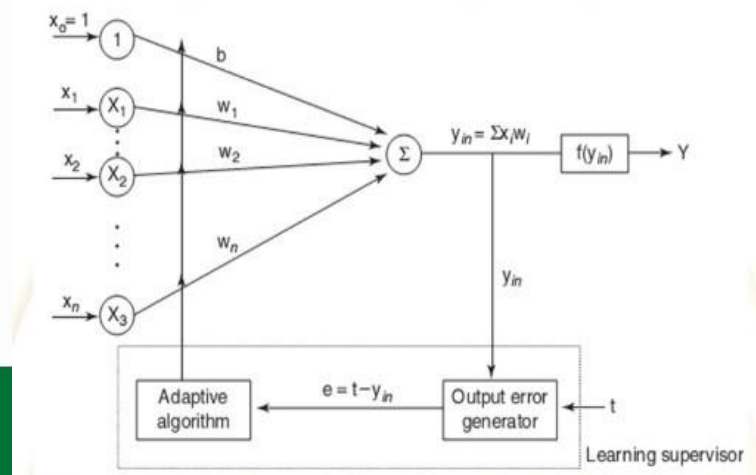
- **Bernard Widrow** and **Marcian Hoff** created two neural network models.
- The model was called **Adaline** (*Adaptive Linear Neuron*), a model that could detect binary patterns and eliminate echo on phone lines.



Bernard Widrow



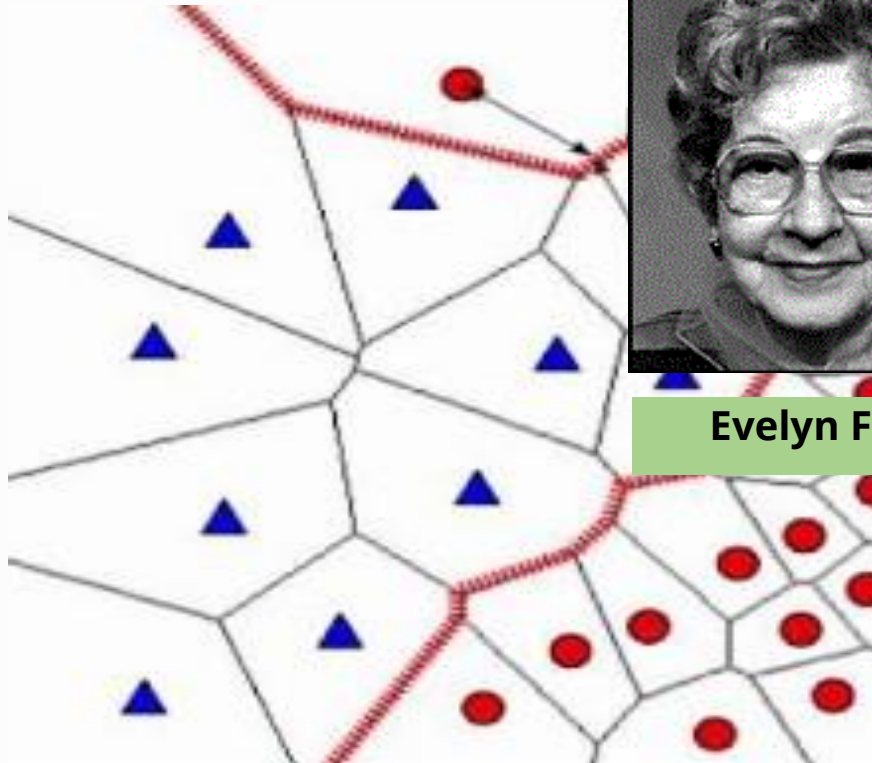
Marcian Hoff



Brief History of Machine Learning (Cont'd)

1951

- **Evelyn Fix** and **Joseph Hodges** developed a non-parametric method for **pattern classification**, which was later expanded by **Thomas Cover**. Later on, this came to be known as the **K-Nearest Neighbor algorithm**.
- The **Nearest Neighbor Algorithm** was written that allowed computers to use very basic pattern recognition.



Evelyn Fix



Joseph Hodges

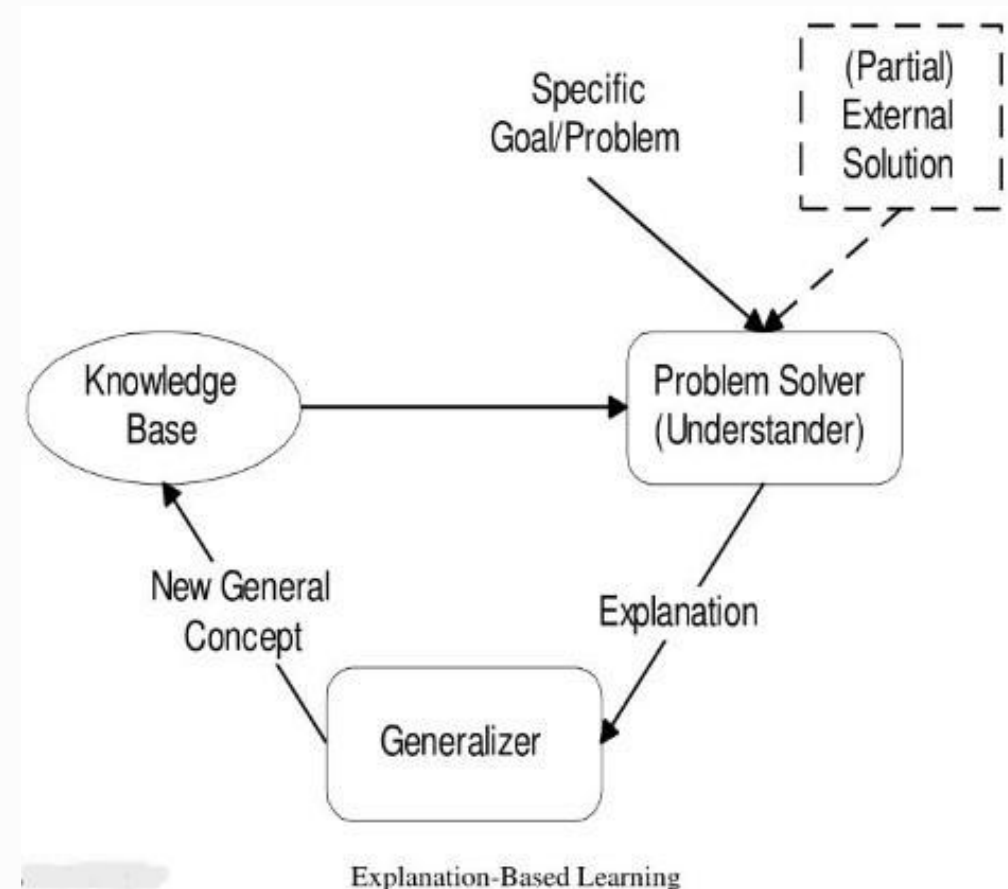


Thomas Cover

Brief History of Machine Learning (Cont'd)

1986

- **Gerald DeJong** introduced the concept of **Explanation-Based Learning (EBL)**, in which a computer analysis data and creates a general rule to discard unimportant information.





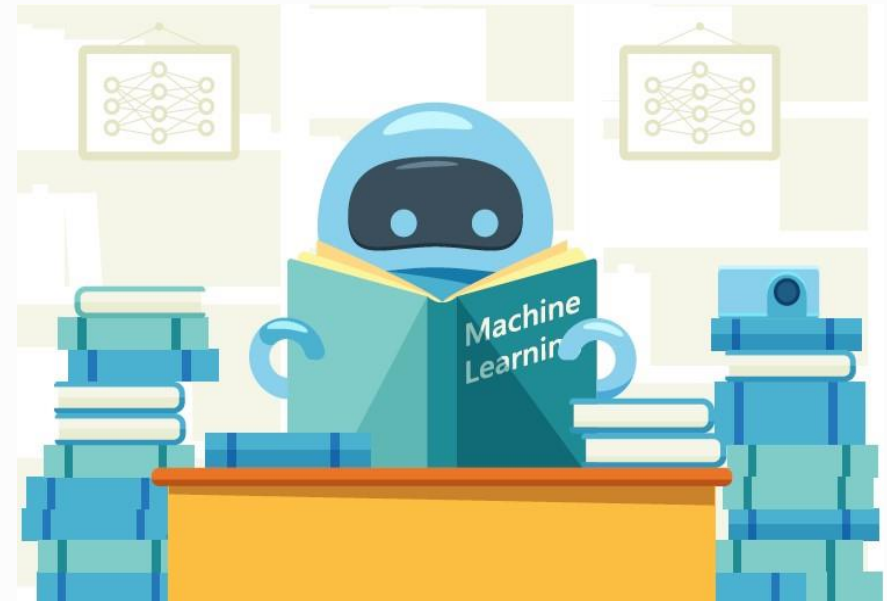
Brief History of Machine Learning (Cont'd)

1990's

- Shifted from a knowledge-driven approach to a more data-driven approach.
- Scientists began creating programs for computers to analyze large amounts of data and draw conclusions or **"learn"** from the results.
- Then, as time goes by, several developments formulated into the modern age of machine learning.

Understanding Machine Learning

- Various sectors of the economy are dealing with huge amounts of data available in different formats from disparate sources.
- The enormous amount of data, known as big data, is becoming easily available and accessible due to the progressive use of technology, specifically advanced computing capabilities and cloud storage.



Understanding Machine Learning

- Companies and governments realize the **huge insights** that can be gained from tapping into big data but **lack the resources and time required** to comb through its wealth of information.
- As such, **artificial intelligence** measures are being employed by different industries to **gather**, **process**, **communicate**, and **share** useful information from data sets.





Terminologies of Machine Learning

- **Model** – a.k.a “*hypothesis*”, a machine learning model is the mathematical representation of a real-world process. A machine learning algorithm along with the training data builds a machine learning model.
- **Feature** - a *measurable property* or *parameter* of the data-set.
- **Feature Vector** - a set of *multiple numeric features*. We use it as an input to the machine learning model for training and prediction purposes.
 - **Example:** predicting housing prices, *feature vector* look like this: [3, 1500, 37.8, -122.4], which 3 bedrooms, 1500 square footage, 37.8 latitude and -122.4 longitude of the house.
- **Training** - An algorithm takes a set of data known as “*training data*” as input. The learning algorithm finds patterns in the input data and **trains the model** for expected results (target). The output of the training process is the machine learning model.



Terminologies of Machine Learning

- **Prediction** – Once the machine learning model is ready, it can be fed with input data to provide a **predicted output**.
- **Target (label)** - The value that the machine learning model has to predict is called the **target** or **label**.
- **Overfitting**- When a massive amount of data trains a machine learning model, it tends to learn from the **noise** and **inaccurate** data entries. Here, the model fails to characterize the data correctly.
- **Underfitting** - It is the scenario when the model fails to decipher the underlying trend in the input data. It destroys the **accuracy** of the machine learning model. In simple terms, the model or the algorithm **does not fit the data well enough**.



Importance of Machine Learning

- It allows companies to transform processes that were previously only possible for humans to perform.

Examples:

- Responding to customer service calls
- Bookkeeping
- Reviewing text/sentences if there are foul words
- Reviewing resumes for everyday businesses.
- Etc.

- Machine learning can also scale to handle larger problems and technical questions.

Examples:

- Predicting natural disaster locations and timelines
- Image detection for self-driving cars



Importance of Machine Learning (Cont'd)

- Machine learning data analysis uses algorithms to continuously improve itself over time, but "quality data" is necessary for these models to operate efficiently.

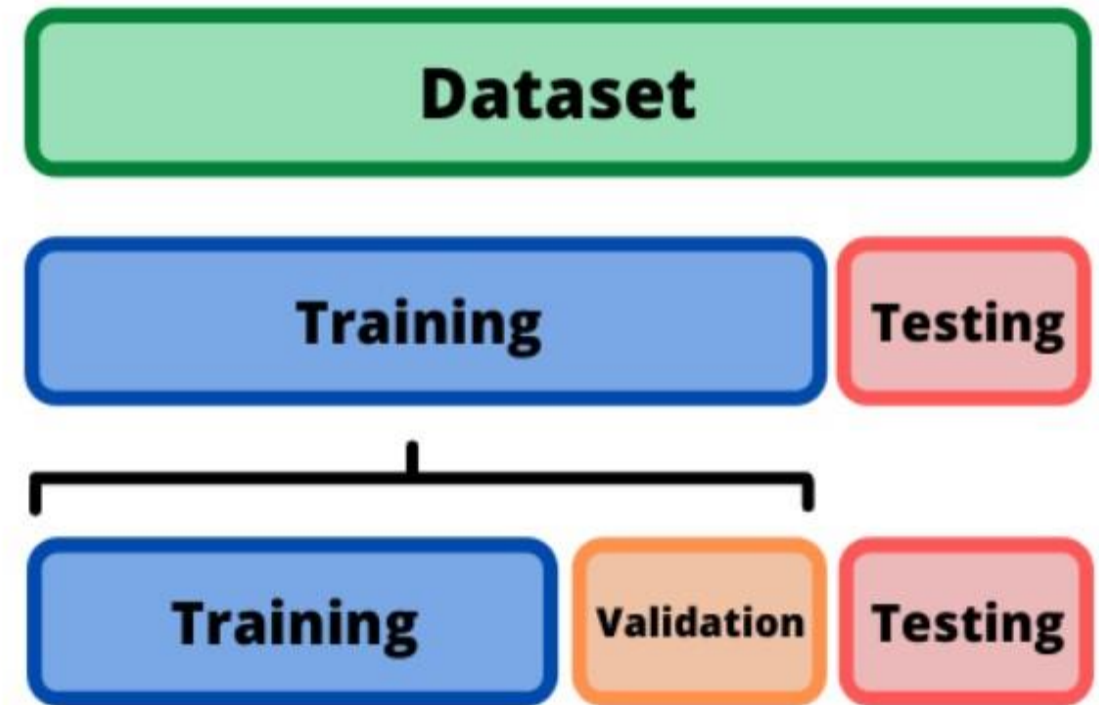
Now, what is Dataset?

- Understanding dataset, we need to know the components of a dataset.
- A single row of data is called an instance.
- Datasets** are a collection of instances that all share a common attribute.
- Machine learning models will generally contain a few different datasets, each used to fulfill various roles in the system.

Importance of Machine Learning (Cont'd)

Training Datasets

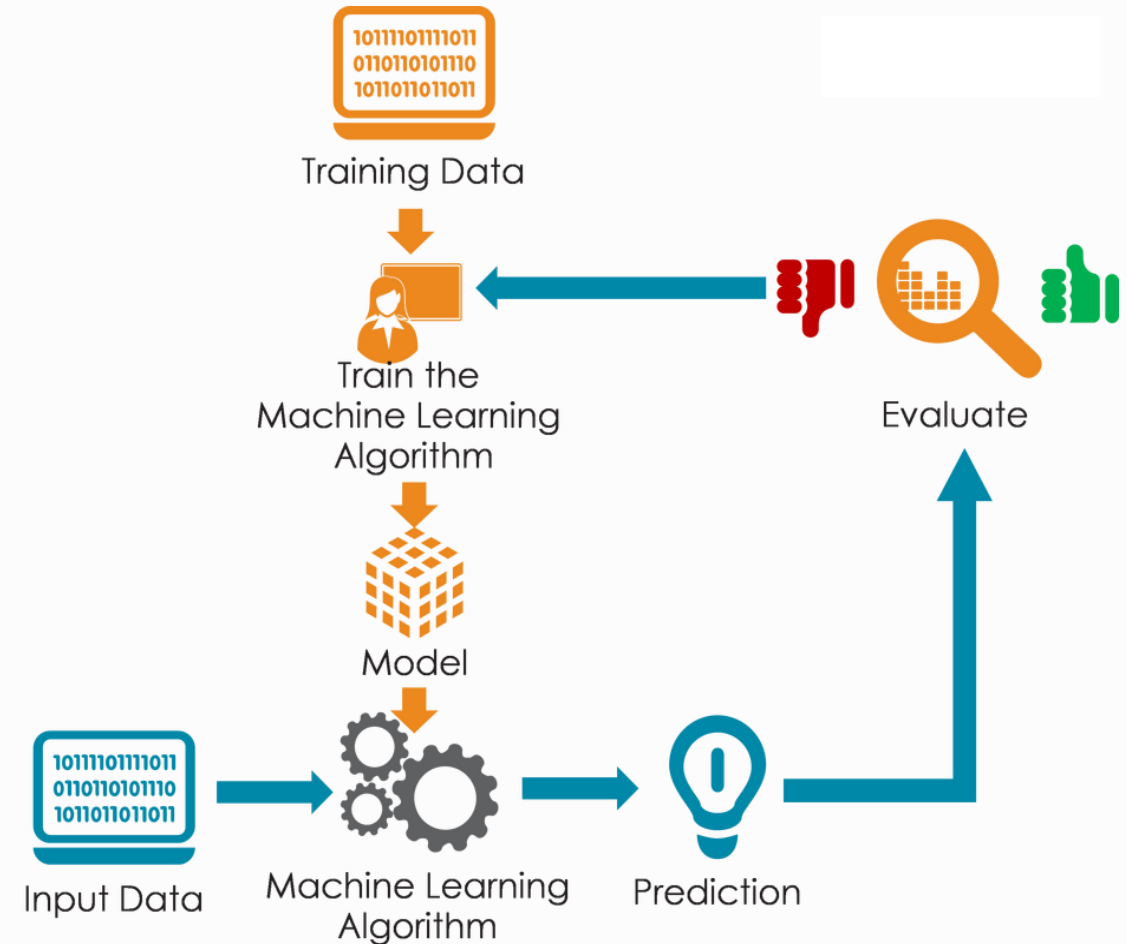
- For machine learning models to understand how to perform various actions, training datasets must first be fed into the machine learning algorithm, followed by validation datasets (or testing datasets) to ensure that the model is interpreting this data accurately.



Importance of Machine Learning (Cont'd)

Training Datasets

- Once you feed these **training** and **validation sets** into the system, subsequent datasets can then be used to sculpt your **machine learning model** going forward.
- The more data you provide to the ML system, the faster that model can **learn** and **improve**.





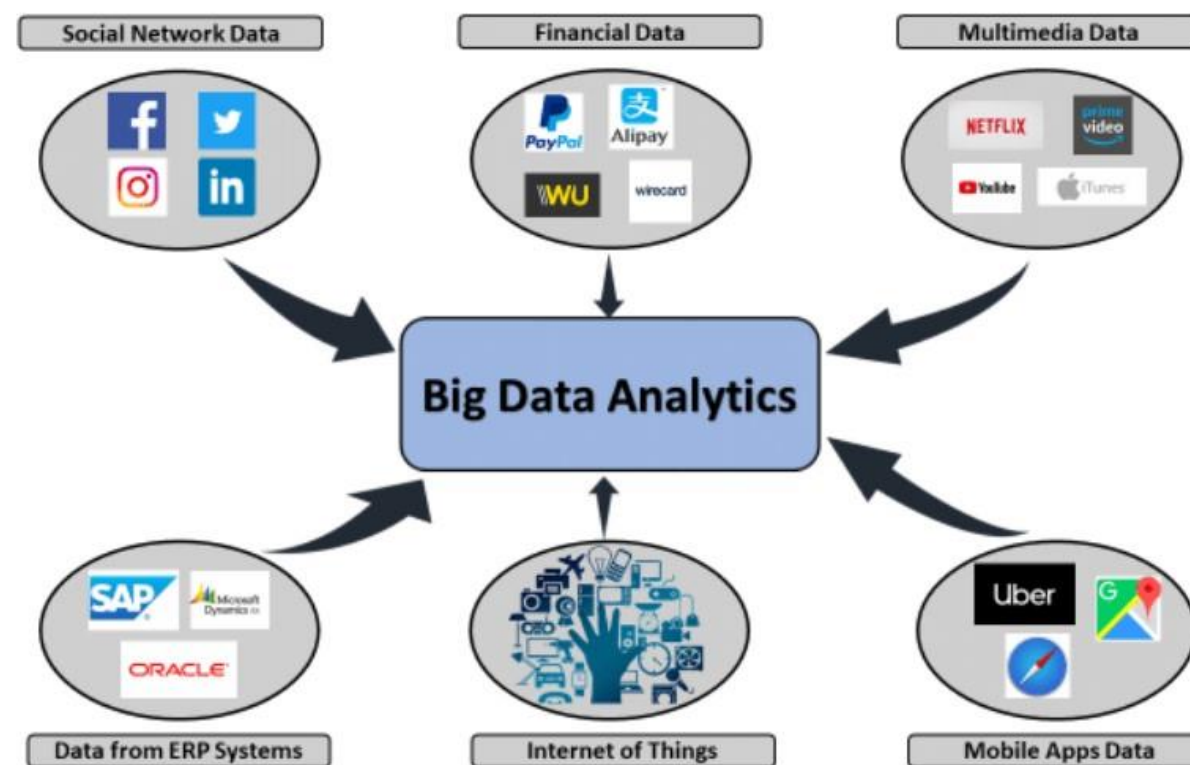
Elements of Machine Learning

- The 6 elements of **Machine Learning** are:
 - ✓ Data
 - ✓ Task
 - ✓ Model
 - ✓ Loss Function
 - ✓ Learning Algorithm
 - ✓ Evaluation

Elements of Machine Learning (Cont'd)

1. Data

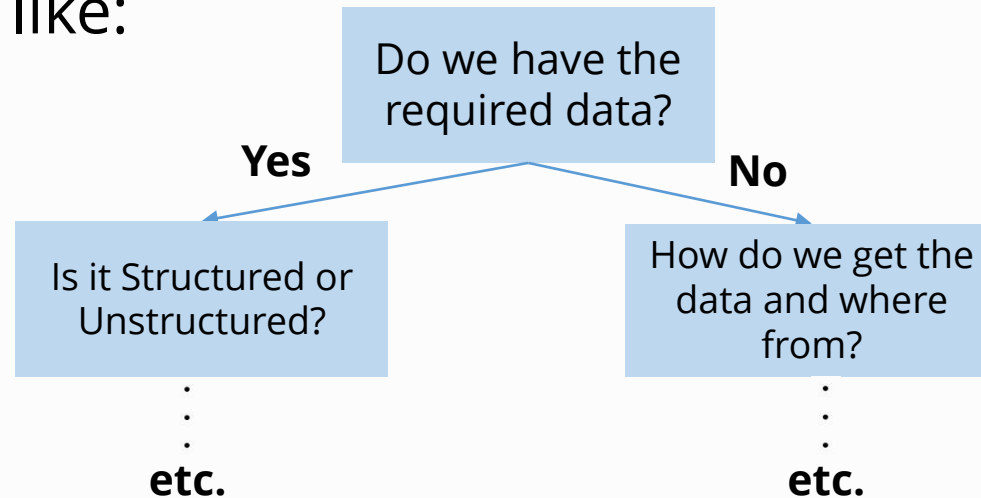
- Data simply means information. All types and formats of information.
- Today, there are enormous amount of data produced every second and this data can be used to answer so many questions.
- **Data** is the fuel of Machine Learning



Elements of Machine Learning (Cont'd)

1. Data

- When starting a Machine Learning project, we need some answers to the questions like:

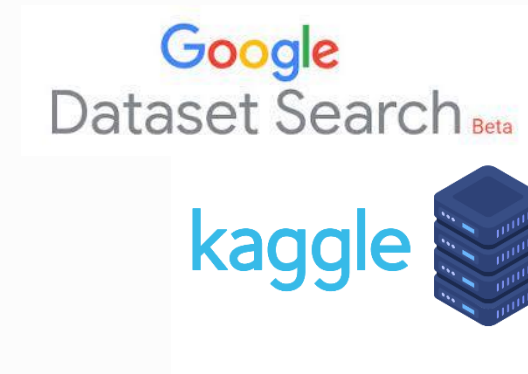


- These questions lead us to different paths in the project.

Elements of Machine Learning (Cont'd)

1. Data

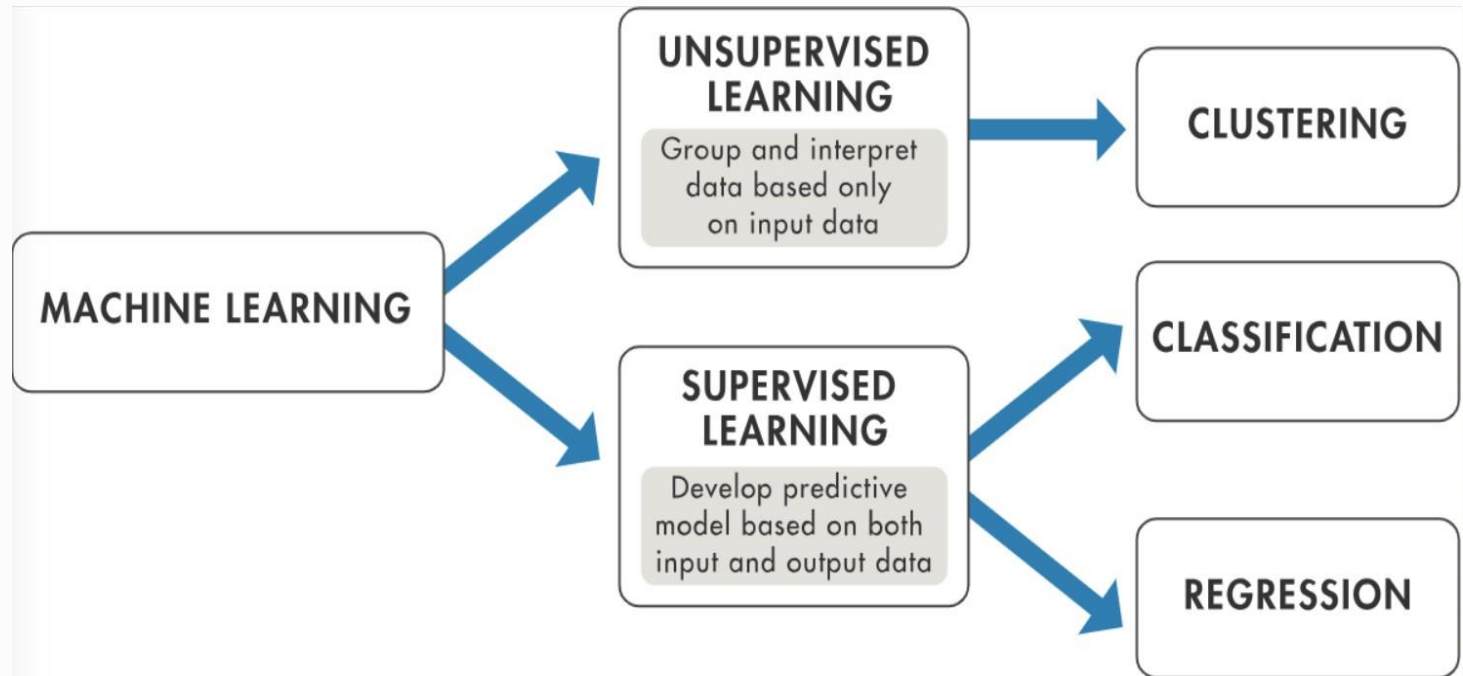
- If we don't have the data needed, then **Data Curation** comes in picture, which is the organization and integration of data collected from various sources.
- If you are lucky enough, there are many resources where you can get datasets that are publicly available:
 - **Google Dataset Search**
 - **Kaggle Datasets**
 - **Open Government Data (OGD)**
 - etc.



Elements of Machine Learning (Cont'd)

2. Task

- After you gathered required data, next question you should ask:
 - "What task do I need to achieve?"
 - "What task can I achieve using the data that I have?"



Elements of Machine Learning (Cont'd)

2. Task

2.1 Supervised Learning

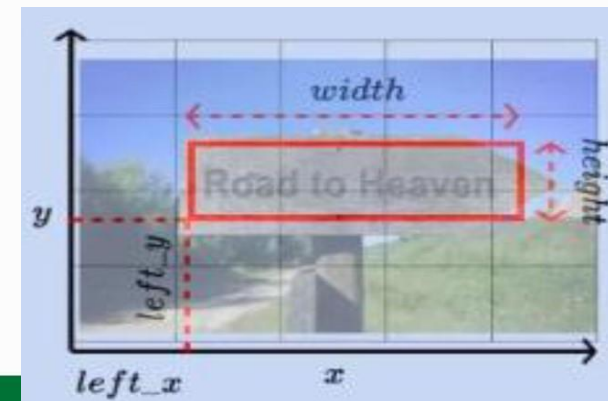
Supervised learning is further divided into **two** sub sections:

- **Regression** – means you want to predict some continuous value, for example: House Price Prediction.
- **Classification** - means you want a discrete answer such as **yes** or **no**. Example: given an image you want to detect whether there is a **dog** or **not** in that image



x	left_x	left_y	width	height
-8.5 -1.7 ... 9.0 7.2	2.3	1.2	9.2	10.1
0.9 -2.1 ... -8.1 1.9	4.3	4.2	7.1	5.1
2.9 -4.5 ... -3.7 8.9	2.3	7.2	6.9	7.3

Supervised



Regression

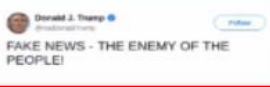
Elements of Machine Learning (Cont'd)

2. Task

2.2 Unsupervised Learning

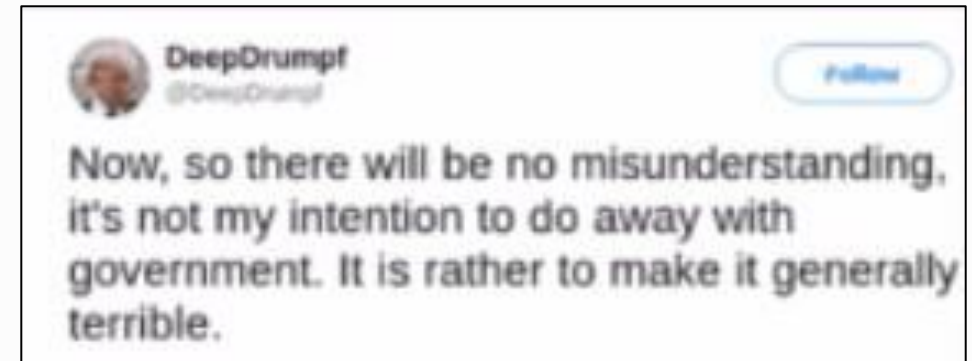
- Often used for Generation and Clustering.
- For example the image given shows that given the tweets of US President Donald Trump as an input, machine generates more such tweets.

Unsupervised

Tweets	x					
 Wow, but not surprisingly, ObamaCare was just ruled UNCONSTITUTIONAL by a highly respected judge in Texas. Great news for America!	2.3	5.9	...	11.0	-0.3	8.9
 For the record, there were MANY people who wanted to be the White House Chief of Staff. Mick M will do a GREAT job!	-8.5	-1.7	...	-1.3	9.0	7.2
 FAKE NEWS - THE ENEMY OF THE PEOPLE!	-0.4	6.7	...	-2.4	4.7	-6.2



Generation



Elements of Machine Learning (Cont'd)

3. Model

- A mathematical function which defines relationship between input data and output data.
- A model can be a simple Linear Function such as $y = mx + c$ or it can be a complex function such as $y = \sin(\cos(ax)) + \tan(\sin(bx^2 + cx))$.
- Always remember, there are many models available that are created over the years and each one has its own pros and cons and each one is suitable for a unique task.

What are the choices for $\hat{f}$?

Models

$$\hat{y} = \hat{f}(x) \text{ [our approximation]}$$

x	y
0.5	14.8
0.2	13.3
0.6	11.6
$\vdots$	$\vdots$
0.3	6.16

$$\hat{y} = mx + c$$

$$\hat{y} = ax^2 + bx + c$$

$$\hat{y} = ax^3 + bx^2 + cx + d$$

$$\hat{y} = ax^4 + bx^3 + cx + d$$

$$\vdots$$

$$\hat{y} = ax^{25} + bx^{24} + \dots + cx + d$$

Elements of Machine Learning (Cont'd)

4. Loss Function

- The loss function is a method of **evaluating** how well your machine learning algorithm **models your featured data set**.
- In other words, loss functions are a **measurement** of how good your model is in terms of predicting the expected outcome.
- This is where we analyze which model **best fit** for estimating the correct relationship for the task.

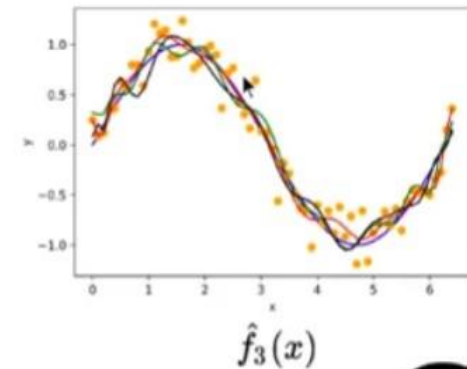
How do we know which model is better?

	x	y	$\hat{f}_1(x)$	$\hat{f}_2(x)$	$\hat{f}_3(x)$
1	0.00	0.24	0.25	0.32	0.08
2	0.10	0.08	0.09	0.30	0.20
3	0.20	0.12	0.11	0.31	0.14
....					
n	6.40	0.36	0.36	0.22	0.15

👤 $\hat{f}_1(x) = 1.79x^{25} - 4.54x^{24} + \dots - 1.48x + 2.48$

👤 $\hat{f}_2(x) = 2.27x^{25} + 9.89x^{24} + \dots + 2.79x + 3.22$

👤 $\hat{f}_3(x) = 3.78x^{25} + 1.57x^{24} + \dots + 1.01x + 8.68$



Elements of Machine Learning (Cont'd)

4. Loss Function

- As for example, the loss function we use is **"Square Error Loss"**. Counting loss function value for all three models, we get to know that the best model for our data is the one with the minimum value of loss function which is **1.38**.
- There are different function loss:
 - ✓ Cross-Entropy Loss
 - ✓ KL divergence
 - ✓ Hinge Loss
 - ✓ Huber Loss etc.
- The lower the loss better the model is in terms of accurate prediction.

$$\begin{aligned}\mathcal{L}_1 &= \sum_{i=1}^n (y_i - \hat{f}_1(x_i))^2 \\ &= (0.24 - 0.25)^2 + (0.08 - 0.09)^2 + (0.12 - 0.11)^2 + \dots + (0.36 - 0.36)^2 \\ &= 1.38\end{aligned}$$

	x	y	$\hat{f}_1(x)$	$\hat{f}_2(x)$	$\hat{f}_3(x)$
1	0.00	0.24	0.25	0.32	0.08
2	0.10	0.08	0.09	0.30	0.20
3	0.20	0.12	0.11	0.31	0.14
....					
n	6.40	0.36	0.36	0.22	0.15



$$\mathcal{L}_1 = \sum_{i=1}^n (y_i - \hat{f}_1(x_i))^2 = 1.38$$



$$\mathcal{L}_2 = \sum_{i=1}^n (y_i - \hat{f}_2(x_i))^2 = 2.02$$



$$\mathcal{L}_3 = \sum_{i=1}^n (y_i - \hat{f}_3(x_i))^2 = 2.34$$

Elements of Machine Learning (Cont'd)

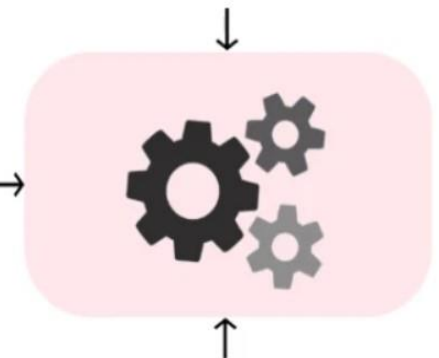
5. Learning Algorithm

- This element will look on how we will estimate different parameters.
- The success of machine learning lies under this element, for parameter estimation in machine learning is a kind of search operation.
- We can compute the parameters through learning algorithm and it becomes an optimization problem where we try to optimize the parameters by minimizing the loss

How do you formulate this mathematically ?

Budget (100crore)	Box Office Collection(100 crore)	Action Scene times (100 mins)	IMDB Rating
0.55	0.66	0.22	4.8
0.68	0.91	0.77	7.2
0.66	0.88	0.67	6.7
⋮	⋮	⋮	⋮
0.72	0.94	0.97	8.1
0.58	0.74	0.35	5.3

$$\hat{f}(x) = ax_1^2 + bx_2^3 + cx_3^2$$



$$\min_{a,b,c} \mathcal{L}_1 = \sum_{i=1}^n (y_i - \hat{f}(x_i))^2$$



Many optimization solvers are available

Elements of Machine Learning (Cont'd)

6. Evaluation

- It is the process of using different **evaluation metrics** to understand a machine learning model's **performance**, as well as its strengths and weaknesses.
- **Model evaluation** aims to estimate the generalization **accuracy** of a model on future.
- **Model evaluation** is important to assess the efficacy of a model during initial research phases, and it also plays a role in model monitoring.

How do we compute a score for our ML model?

Evaluation

Top - 1

	True Labels	Predicted Labels	
	[2.1, 1.2, ..., 5.6, 7.8]	1	1 ✓
	[3.5, 6.6, ..., 2.5, 6.3]	2	3 ✗
	[6.3, 2.6, ..., 4.5, 3.8]	3	1 ✗
	[2.8, 3.6, ..., 7.5, 2.1]	4	4 ✓
	[2.2, 1.7, ..., 2.5, 1.8]	5	5 ✓
	[6.3, 2.6, ..., 4.5, 3.8]	3	2 ✗
	[1.9, 3.3, ..., 4.2, 1.1]	5	5 ✓

$$Accuracy = \frac{\text{Number of correct predictions}}{\text{Total number of predictions}}$$
$$= \frac{4}{7} = 0.55$$

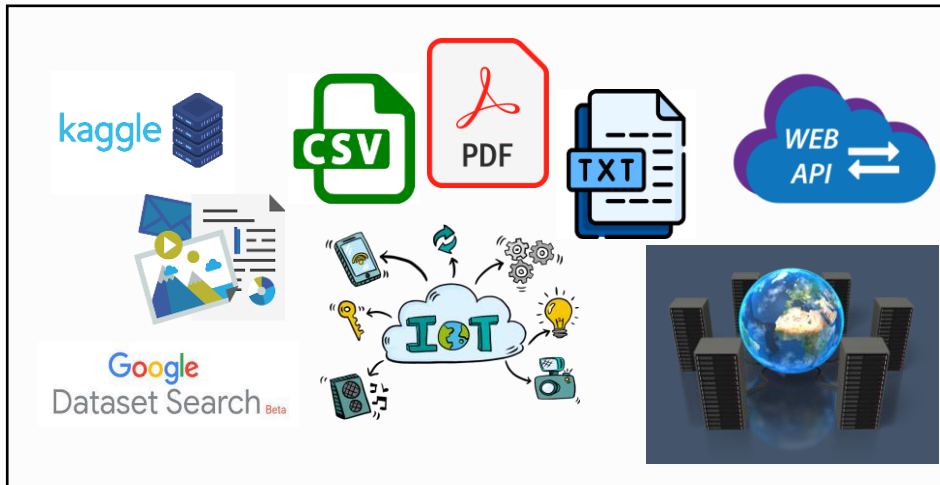
Class Labels	
Lion	1
Tiger	2
Cat	3
Giraffe	4
Dog	5



Machine Learning Process

1. Data Collection

- Collection / gathering data from various data sources.
- The quantity & quality of your data dictate how accurate our model is.
- The outcome of this step is generally a representation of data (Guo simplifies to specifying a table) which we will use for training.
- Using pre-collected data, by way of datasets from Kaggle, UCI, etc., still fits into this step.



Machine Learning Process (Cont'd)

2. Data Preparation

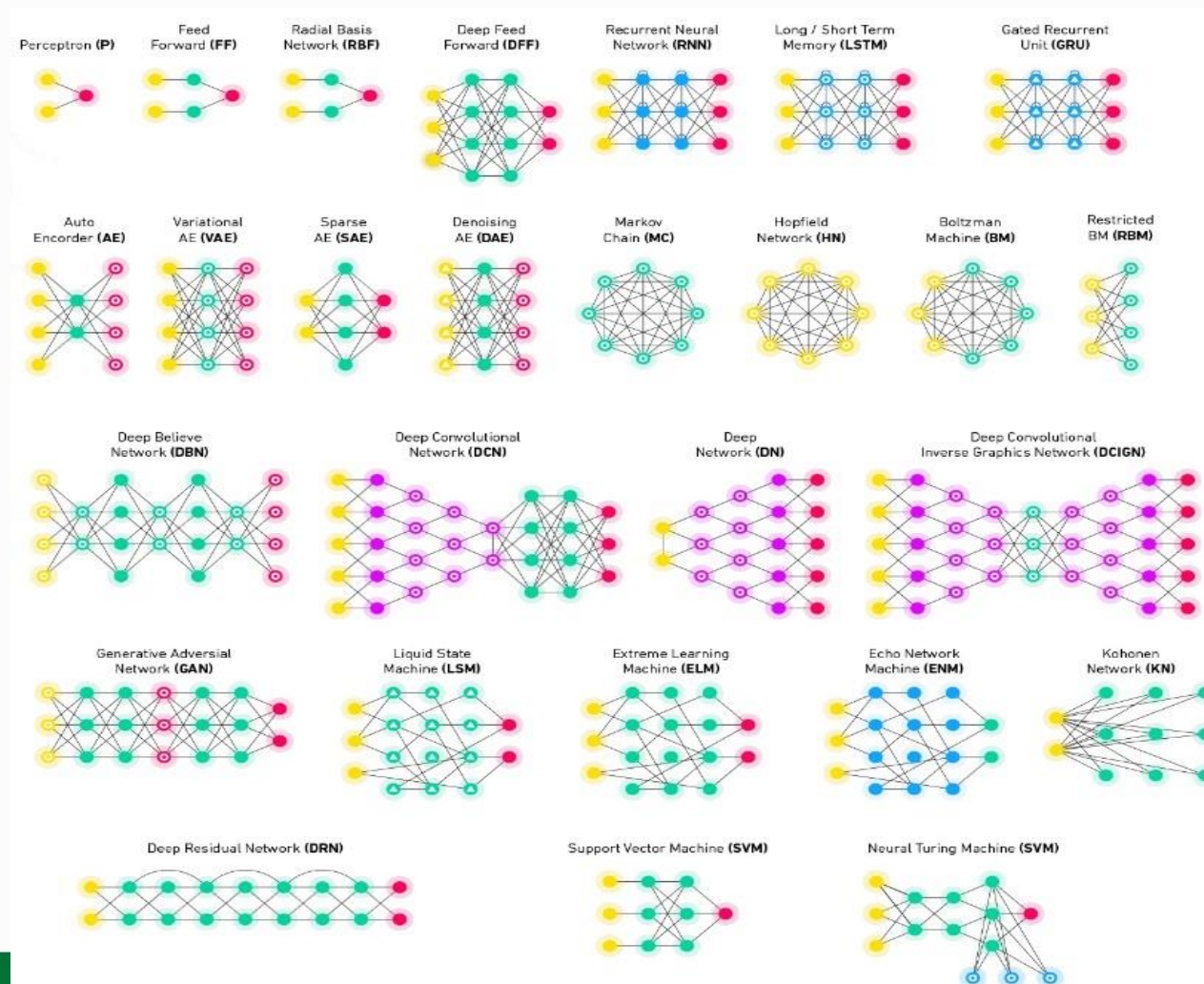
- Wrangle data and prepare it for training.
- **Clean** the data which may require removing duplicates, correcting errors, deal with missing values, data normalization, data type conversions, **etc.**).
- Randomize data, which erases the effects of the particular order in which we collected and/or otherwise **prepared our data**.
- **Visualize data** to help detect relevant relationships between variables or class imbalances (**bias alert!**), or perform other exploratory analysis.
- Split into training and evaluation sets.



Machine Learning Process (Cont'd)

3. Choose a Model

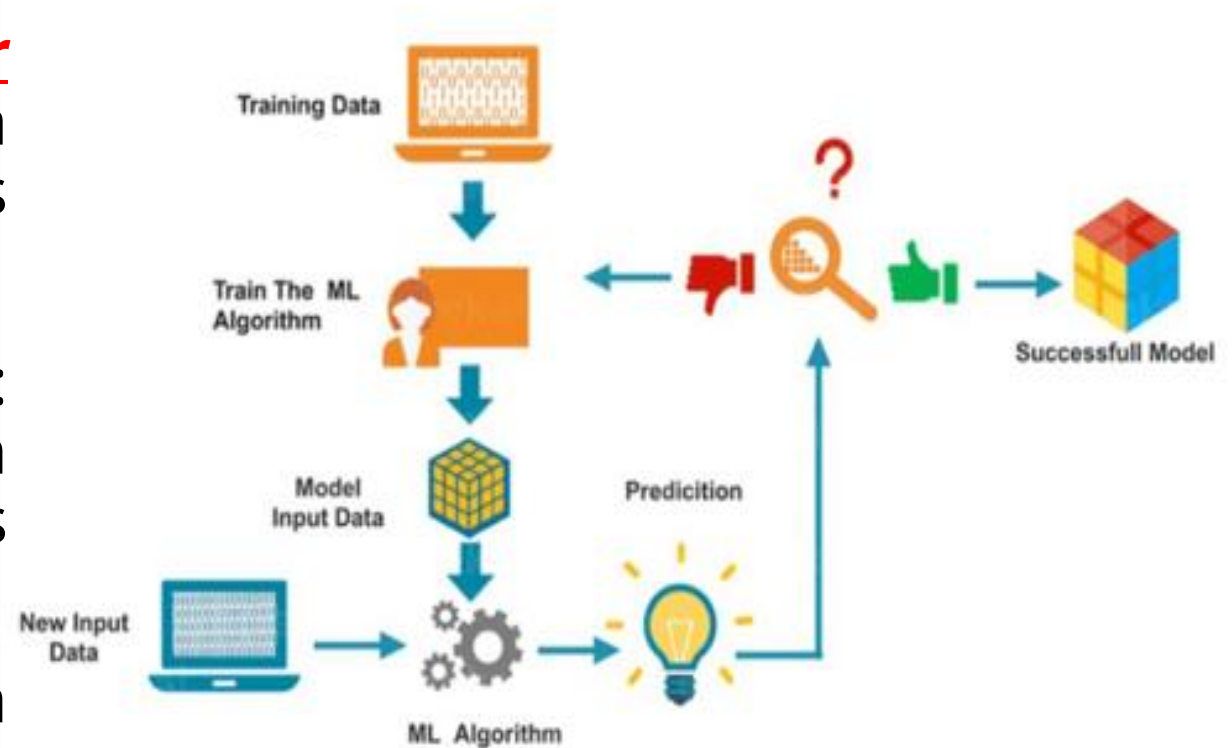
- There are different algorithms for different tasks.
- Each algorithms has its uniqueness. Choose the right one.



Machine Learning Process (Cont'd)

4. Train the Model

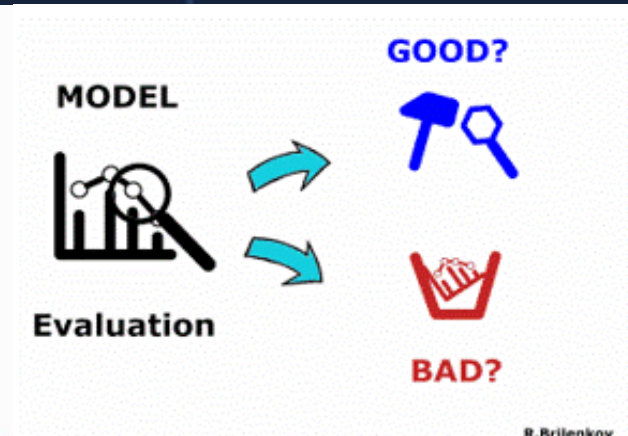
- The goal of training is to answer a question or make a prediction correctly as often as possible.
- Linear regression example: algorithm would need to learn values for m (or W) and b (x is input, y is output).
- Each iteration of process is a training step



Machine Learning Process (Cont'd)

5. Evaluate the Model

- Uses some metric or combination of metrics to "measure" objective performance of model.
- Test the model against previously unseen data.
- This unseen data is meant to be somewhat representative of model performance in the real world, but still helps tune the model (*as opposed to test data, which does not*)
- Good train/eval split? 80/20, 70/30, or similar, depending on domain, data availability, dataset particulars, etc.



R.Brilenkov



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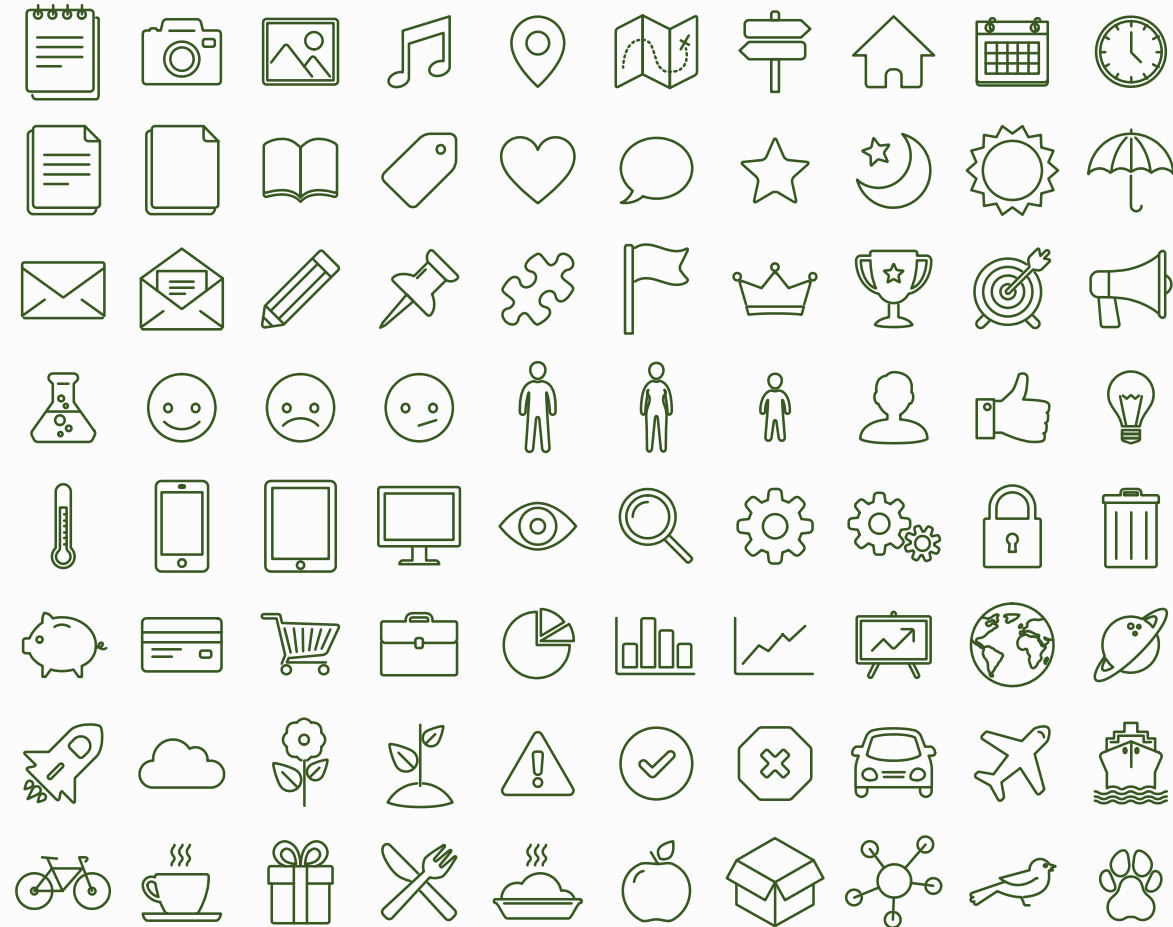
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